

Research Article

Precarious Data: Crack, Opioids, and Enacting a Social Justice Ethic in Data Visualization Practice

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Abstract—Background: The linguistic framing strategies used in media reporting on illegal drugs have been extensively documented, but less attention has been directed toward visuals, particularly data visualizations.

Literature review: Positioning illegal drug use as a criminal justice problem or a public health issue are types of frameworks that use specific rhetorical strategies. **Research questions:** 1. What are the rhetorical strategies used in data visualizations published during the crack and opioid drug epidemics, respectively? 2. Do these strategies advance dominant media narratives that crack addiction should be criminalized but opioid addiction should be treated like a public health issue? And if so, how is this accomplished? **Methodology:** Drawing from the media studies approach previously employed in a study in technical and professional communication (TPC) on information design trends, I apply the concept of “scripto-visual” rhetoric to select data visualizations published by mainstream news media during both drug epidemics. **Results:** I argue these graphics escalated the perceived threat during both drug epidemics but different scripto-visual rhetorical strategies were used. **Conclusions:** Attending to ethical considerations in the creation of data visualizations has long been important in TPC, while scholarship has integrated social justice as a core component of the discipline. In the last section of this article, I bring these themes together by arguing that a social justice ethic is needed in data design work. I then propose a critical heuristic constructed from Jones et al.’s positionality, privilege, and power framework that can be used analytically or as an intentional tool to tease out the ways particular scripto-visual rhetorical decisions may be promoting inequities.

Index Terms—Data visualizations, drug epidemics, news media, rhetorical analysis, social justice.

In late 2015, two economists—Anne Case and Angus Deaton—were studying mortality data when they discovered an alarming trend: death rates for middle-aged, White Americans were dramatically increasing [1]. Their findings brought national media attention to the opioid epidemic [2], which has since become a national public health priority (see, e.g., [3]–[5]). This dramatic response, as critics have argued, stands in stark contrast to an arguably equally devastating drug epidemic that hit many US cities several decades ago: the influx of crack cocaine (see, e.g., [6], [7]). Indeed, people addicted to opioids, who tend to be White, have been positioned as patients needing treatment, whereas those addicted to crack, who tended to be people of color, were positioned as criminals [8].

Media coverage influences how social issues are perceived by nonexpert audiences [9], [10] and the decisions that are made to address these issues

[11], [12]. The reporting strategies used to frame illegal drugs are particularly important because these choices determine what solutions will be emphasized—that is, “increased law enforcement ... [or] job creation, drug treatment, or educational programs” [11, p. 427], or “public policy responses” [8, p. 214]. The profoundly deleterious effects of laws enacted during the crack epidemic (e.g., Anti-Drug Abuse Act of 1986) that disproportionately penalized people of color can still be seen decades later [13]–[15]. The linguistic framing strategies used in reporting on illegal drugs have been extensively documented (e.g., [16]–[21]). However, less attention has been directed toward visuals (see [22]–[25] for exceptions), particularly data visualizations, which are now routinely used in news coverage.

Drawing from the media studies approach previously employed in a study in technical and professional communication (TPC) on information design trends [26], in this article, I applied the concept of “scripto-visual” rhetoric—a term coined by Dawkins to describe the “effects of images ... [resulting] from the interaction between pictorial structures and language” [27, p. 81]—in analyzing selected data visualizations published by mainstream news media during the crack and

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Practitioner Takeaway

- Data visualizations published in mainstream media coverage of the crack and opioid epidemics, respectively, escalated the perceived threat during both of these drug epidemics.
 - However, different scripto-visual rhetorical strategies—that is, genre choices and subject matters—were used in these graphics to reinforce dominant media narratives that positioned crack addiction as a crime but opioid addiction as a public health issue.
 - Bar charts showing crime-related information were used during the crack epidemic to emphasize increases between years, promoting the need for immediate law enforcement solutions. In contrast, line graphs showing mortality and health-related information were used during the opioid epidemic to position addiction to these drugs as a long-term public health issue. These results demonstrate that a social justice ethic is needed in data design work.
 - A critical heuristic constructed from Jones et al.'s positionality, privilege, and power framework can be used analytically or as an inventional tool to tease out the ways that particular scripto-visual rhetorical decisions may promote inequities.
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opioid drug epidemics, respectively. More specifically, I investigate the way that text and image work together in this context to “purposeful[ly] and rhetorically ... motivate belief or action,” as Ross [28, p. 148] characterized Dawkins’ idea in his work on scientific illustration and ethics in the field of TPC.

Indeed, data visualizations rely on both visual and linguistic/textual semiotic modes to construct meaning. More specifically, the genre that a creator chooses (e.g., bar chart, line graph) communicates the visual relationships that will be emphasized (e.g., comparisons, trends over time) whereas the textual content (e.g., titles, figure labels, captions) communicates the subject matter of these relationships. In analyzing the dataset that I created for this study, I focus on the visual message conveyed through genre and the textual message conveyed through the titles of select graphics, as explained in more detail in the Methodology section.

I argue that graphics published during both drug epidemics emphasized “incredible numbers,” a reporting strategy used during outbreaks [29, pp. 483–484]—epidemics in the historic meaning of the word [30]—to escalate the perceived threat. However, I propose that different scripto-visual rhetorical strategies were used to reinforce dominant media narratives that positioned crack addiction as a crime but opioid addiction as a public health issue. More specifically, bar charts showing crime-related information were used during the crack epidemic to emphasize increases between years, promoting the need for immediate law enforcement solutions. In contrast, line graphs showing mortality and health-related information were used

during the opioid epidemic to position addiction to these drugs as a long-term public health issue.

Attending to ethical considerations in the creation of data visualizations has long been important in TPC (see, e.g., [31]–[33]), whereas more recently Jones et al. have argued that our field’s broad emphasis on ethics can be connected to adopting an active “social justice stance” [34, p. 211]. In the last section of this article, I bring these themes together to argue that the results of this study demonstrate that a social justice ethic is needed in data design work. I then show how this ethic can be enacted—that is, put into practice—by drawing from Jones et al.’s positionality, privilege, and power framework [34] (see also [35]), which “coconstruct[s] the ways in which people engage with identity markers...” [34, p. 220], to propose a critical heuristic that can be used analytically or as an inventional tool to tease out the ways particular scripto-visual rhetorical decisions—genre choice and subject matter—may be promoting inequities. I conclude by illustrating how this heuristic can be applied analytically by evaluating one of the graphics published during the crack epidemic.

Visualizing data involves making scripto-visual rhetorical choices that shape the way visual quantitative information is used to construct arguments. This study describes how these strategies were used in a particular high-stakes communicative context (i.e., media coverage of drug epidemics) to position some people as worthy of protection (e.g., White people addicted to opioids) but left others vulnerable (e.g., people of color addicted to crack)—like the precarious data visualizations created during the crack epidemic.

LITERATURE REVIEW: MEDIA FRAMEWORKS USED DURING DRUG EPIDEMICS

The notion that media coverage frames social issues in ways that shape public perception is well-established in media studies. The concept of frameworks, which originated in Goffman's [36] work on frame analysis, "essentially involve[s] *selection* and *salience*" as media and public affairs scholar Entman put it in his foundational definition [9, p. 52; italics in original]. More specifically, media frameworks are "overarching master narrative[s]" [37, p. 251] that convey "what will be discussed, how [the topic] will be discussed, and above all, how it will not be discussed," as Altheide put it [38, p. 651]. Positioning illegal drug use as a criminal justice problem or as a public health issue, for instance, are types of frameworks that use specific rhetorical strategies; the former employs punitive language and emphasizes law enforcement actions, whereas the latter focuses on "treatment, intervention, and even education" [39, p. 53].

To illustrate, Beckett argued that coverage during the war on drugs relied on government and law enforcement sources who emphasized the situation as a "law and order" issue [40, p. 163]. In this study, she also identified themes using keywords and short phrases that she proposed were intended to construct particular "drug issue packages" [pp. 169–174] like "get tough" and "zero tolerance" [pp. 170–171]. More than 20 years later, Orsini found similar categories in her content analysis of reporting on cocaine and heroin from 2000 to 2015 [17]. More specifically, she argued that "traditional prohibition-oriented [policy] approaches" [p. 204] continued to be emphasized, as well as the use of "social control frames" [p. 205] through themes she characterized as *ongoing fight*, *dangerous use*, *violent traffickers*, and *fallen star*.

In some cases, reporting about a specific drug can dramatically escalate (e.g., crack cocaine in the late 1980s, methamphetamine in the late 1990s/early 2000s [20], and mephedrone in the UK in 2010 [16], leading to a moral panic [41]. During such "media-generated exaggeration[s]" [42, p. 119], "drug-scare rhetoric" [20, p. 227]—for example, referring to the situation as a "plague" [20] or an "epidemic"—is often used to reinforce "the *urgency* of swift and serious action," criminologist Chiricos [43, p. 117, italics in original] argued, which can then be used to "justify expanding the punitive apparatus of the state" [p. 104]. Indeed, rhetorically, the term *epidemic* invokes "the fear and panic that traditionally accompanied

life-threatening infectious diseases ..." [30, p. 33], and it has been used by the media since the late 20th century "to qualify anything that adversely affects a large number of persons or objects and propagates like a disease, such as crack cocaine ..." [44, para. 16]. The word, too, has been used as its own framing strategy to intensify the perceived threat posed by emergent illegal drugs like the "bath salts" reportedly ingested by the perpetrator in the infamous and highly dramatized "Miami zombie attack" [19].

Both crack and the current opioid situation have been described as "drug epidemics." Yet as Shachar et al. found in their study, they have been framed very differently [8]. More specifically, news articles published during the crack epidemic predominantly used keyword combinations that emphasized punitive rhetoric such as "law+enforcement" and "substances+crime," whereas coverage of opioids used terms like "health," "health+care," and "public+health" [p. 218]. The current opioid situation includes both the legal and illegal use of these drugs [45]. Yet interestingly, from 1998 to 2012, the abuse of prescription opioids was also positioned as a criminal justice issue [46]. The recent shift to a public health framework, Shachar et al. argued, occurred when overdose rates among White Americans eclipsed those of other races [8].

Unfortunately, such racially motivated framing strategies have not been uncommon [47], [48]. For instance, in the late 19th century, newspapers reported that Chinese men who had immigrated to the US to work on the transcontinental railroad were luring White women into opium dens to become sex slaves, leading to the passage of laws that prohibited smoking opium [49], [50]. Further and prior to the crack epidemic, Black Americans had been demonized for cocaine use during the early 20th century [47], [48].

In contrast, scholars have pointed out that White drug users have frequently been portrayed in ways that engender compassion. For example, journalist and media critic Szalavitz claimed that an "innocent victim narrative" has been used to position White people's addiction to opioids as starting in a "doctor's office" rather than "on the streets" [51, para. 10]. Furthermore, in media coverage of heroin and nonprescription opioid use from 2001 to 2010, Netherland and Hansen found that the addiction of White users was positioned as "surprising and novel" [23, p. 672], [48]. White users' substance abuse experiences, they continue,

were represented as “personal, individualized tragedy” in contrast to the “impersonal arrest reports that characterized media accounts of drug use in Black and Latino communities” [p. 675]. For instance, in references to “urban opioid use,” punitive measures were emphasized, whereas accounts of “suburban and rural prescription drug use” advocated “prevention, education, [and] treatment...” [p. 677].

Part of Netherland and Hansen’s discussion focused on the images of White drug users in many of these stories, which they argued push back against longstanding media stereotypes [23]. The White women who are shown, Hansen argued in a different article, visually “convey blamelessness and vulnerability ... challeng[ing] the historically punitive and criminalizing ethos of US drug policy...” [22, p. 322]. As these examples demonstrate, imagery, too, is a central component of media framing, which subtly shapes perception in powerful ways [37], [52].

At the same time, the influence of visuals is not limited to photographs, as Orcutt and Turner’s study on select data visualizations published during the crack epidemic illustrates [25]. More specifically, they argued that several news articles used graphics constructed from misleading numeric information to amplify the perceived threat of “a coke plague” [pp. 194–196]. Fortunately, such practices were not necessarily routine [53]. Yet their study points to the crucial role that data visualizations play in advancing a particular narrative in news coverage.

RESEARCH QUESTIONS

In this article, I explore the following two research questions.

RQ1. What are the scripto-visual rhetorical strategies used in data visualizations published during the crack and opioid drug epidemics, respectively?

RQ2. Do these strategies advance dominant media narratives that crack addiction should be criminalized but opioid addiction should be treated like a public health issue? And, if so, how is this accomplished?

METHODOLOGY

Creating the Dataset: Inclusion/Exclusion Criteria

To investigate these questions, I

searched for graphics in the following major, mainstream news outlets: *The Los Angeles Times*, *The New York Times*, and *The Washington Post*. In choosing these publications, I sought to keep the scope of the dataset manageable but also to include graphics created by news sources that were widely circulating during *both* drug epidemics. More specifically, these newspapers are currently among those with the highest circulation rates [54], and all have directed significant attention to the opioid epidemic. To illustrate, a search for “opioid epidemic” in *The Los Angeles Times* and *The Washington Post* resulted in 444 and 1,739 hits, respectively [55]. During the crack epidemic, *The New York Times* assigned a full-time reporter to cover the situation [56], whereas *The Washington Post* published 1,565 reports about illegal drugs just between October 1988 and October 1989 [57, p. 21]. Finally, although crack may have appeared simultaneously in multiple US cities, *The Los Angeles Times* was the first newspaper to report about the potent new substance that had begun wreaking havoc in South Central Los Angeles—an area where the epidemic would become increasingly severe [58].

In compiling the dataset, I searched for “crack cocaine” and “chart” from January 1, 1986 to December 31, 1992. (During this time, the word “graphic” was used to refer to photographs, and “chart” was used to refer to data visualizations.) I chose this timeframe to align with the period that Reinerman and Levine, two sociologists who have extensively studied the crack epidemic, have proposed constitutes the “anticrack [media] frenzy” [57, p. 8]. I then searched for “opioid” from November 2, 2015—the date *The New York Times* published their first report on Case and Deaton’s landmark study [2]—to June 30, 2018, the date I ceased data collection and began my analysis. These were the only search terms used. I stopped data collection at this point to ensure that the opioid epidemic was not over-represented as I had collected 118 data visualizations of crack and 140 data visualizations of opioids that met the inclusion criteria. Indeed, data visualizations are routinely published in newspapers today, which was not the case in the 1980s [59].

Graphics published in broadcast news reports and reprints were excluded, as well as interactive visualizations published during the opioid epidemic. All visuals that met the initial inclusion criteria were reviewed to ensure that the crack or opioid epidemic, respectively, was the main subject

matter or a dominant theme of the article in which the graphic was published.

Investigating the intentions of the creators who designed these graphics and the extent to which they may have planned to visually frame crack and opioids, respectively, in particular ways is beyond the scope of this study. However, it is important to note that data visualizations published in major news outlets today—like those created during the opioid epidemic—are usually designed by professional visual journalists who collaborate with reporters to jointly plan the visual content. In contrast, during the crack epidemic, some news stories may have used or reproduced graphics from government organizations, whereas others were created by professional artists or graphic designers.

The graphics in this study were coded chronologically by epidemic (crack and then opioids) and publication in alphabetic order using the following abbreviations: *LAT*, *NYT*, and *WAPO*. Each graphic was then assigned a unique tracking number. To illustrate, *LAT2* [60] was published in *The Los Angeles Times* during the crack epidemic and is the second complete visual in the dataset; *WAPO175* [61] was published in *The Washington Post* during the opioid epidemic and is the last. Visuals that included more than one independent graphic, a common practice during the crack epidemic, were also labeled alphabetically as A, B, C, etc. For instance, *LAT1* [62] includes seven unique, stand-alone visuals published as a single unit—three bar graphs, two tables, and two area line graphs—and was coded as *LAT1A–G*. A total of 258 individual images was included in the dataset: 118 for crack (*LAT1* through *WAPO57*) and 140 for opioids (*LAT58* through *WAPO175*). The graphics that comprise the final dataset were selected from 44 articles published during the crack epidemic and 60 articles published during the opioid epidemic.

Data Analysis Approach Like other scientific visuals [63], data visualizations advance arguments that propose that quantitative information should be interpreted in a particular way. In my analysis, I focused on the visual message conveyed through genre and the linguistic/textual message conveyed through themes in the title of each graphic. To complete the first categorization task (i.e., genre analysis), I worked with an expert in biostatistics to classify each graphic in the dataset using a heuristic developed by data visualization expert Ann Emery [64]. Using Emery's rubric, we classified all the graphics independently. We then discussed graphics where there was disagreement

and came to a consensus about the category. Of Emery's 46 genre classifications, we determined that six categories were applicable. Table I includes definitions that are based on Emery's rubric and specific examples from the dataset to illustrate. *Other* was assigned to the small number of graphics (described in the results) that did not fall into one of these six main categories.

Genre classification (using Emery's rubric) provided an established framework for thematically categorizing the visual message of graphics in the dataset. However, because there was not an established rubric that could be used to complete the textual analysis portion, I used a modified grounded theory approach [65], [66]. This methodology has previously been used in TPC studies focusing on visual communication (e.g., [26], [67], [68]), and it involves using an iterative coding process to inductively tease out subject matter themes.

Following Creswell's first step of this methodology, which includes "generating categories of information (open coding)" [69, p. 196], my research assistant and I first jointly open coded the dataset using the keyword combinations (e.g., "law+enforcement," "substances+crime," "health+care," "public+health") that Shachar et al. [8, p. 218] developed for their study. During this first phase, we constantly compared the categories that we were currently assigning to those that we had already assigned [65], discussing similarities and differences, defining our terms in greater detail, recoding graphics that we had previously coded as needed, and creating new classifications.

To illustrate, many of Shachar et al.'s crime-related terms proved to be applicable. However, "substance+abuse+treatment" was the only relevant term from their list of "medical terminology" [8, p. 218]. Thus, we also created several new categories (e.g., *drug use and/or abuse; mortality*) as well as *not enough info* for graphics for which the subject matter could not be determined from the title alone. The level of detail in Shachar et al.'s keyword combinations was important during the first phase of coding because it allowed us to discuss the subject matter of each graphic at length as needed. Indeed, classifying some graphics was straightforward, whereas we discussed others in greater detail to come to consensus about the category.

During the second phase of coding and aligning with Creswell's [69] next step of "positioning [categories] within a theoretical model (axial

TABLE I
GENRE CATEGORIES

Genre	Definition and Examples
Bar charts	Compare continuous quantities among different variables (e.g., NYT10 [103]; LAT58 [104]).
Line graphs (all)	Compare one or more variables over time. These visuals were further classified as follows: • <i>Traditional</i>: Show a trend (three or more consecutive years) spanning a decade or more and use proportional labeling and spacing (e.g., NYT13A [82]; LAT60 [77]). • <i>Area</i>: Same as above; the space between the line and the x-axis is filled to emphasize changes in quantity (e.g., WAPO36B [105]). • <i>Truncated*</i>: Use shorter lines, which gives a shortened or “truncated” appearance; labels are only included for the first and last dates (opioid epidemic only; e.g., NYT64 [80]).
Maps	Show spatial relationships between and among variables. These visuals were further classified as follows: • <i>Choropleth</i>: Use shading or patterning to show the density of particular variable(s) (e.g., NYT26 [106]; LAT61 [107]). • <i>Directional</i>: Include navigational features like roads (crack epidemic only; e.g., NYT12 [108]). • <i>Other</i>: Neither of the above (e.g., NYT76 [86]).
Pictograms and Pie charts	Show a parts-to-whole relationship (e.g., LAT3B [60]; WAPO 114A [109]).
Slope**	Use a straight line to show before/after comparisons between two years (opioid epidemic only; e.g., WAPO99A [81]).
Tables***	Use a grid to show numeric comparisons (e.g., LAT1D [62]; NYT71A [110]).

*I developed this term to describe this type of line graph.

**Slope graphs were coded as a separate category per Emery’s rubric. Unlike truncated line graphs, which show the details of a trend over time like traditional line graphs (i.e., peaks, dips, and/or other curvatures), slope graphs connect two points in time using a straight line.

***Tables are a visual organizational strategy but are generally not considered “graphics” [111].

coding)” (p. 196), we sought to condense the categories that we had developed during open coding into broader themes to determine whether they reflected a criminal justice or a public health framework, respectively, or neither. More specifically, as previously stated, the former uses punitive rhetoric and emphasizes law enforcement actions, whereas the latter focuses on “treatment, intervention and even education” [39, p. 53] as well as health-related themes [8].

To illustrate, we categorized graphics that had previously been classified using criminal justice terms [8] as *crime-related*, and those that we had initially classified as “substance+abuse+treatment” as *health-related*. *Mortality*, which was used to describe graphics that showed the number of deaths, and *not enough info* were created during open coding and became final categories during this phase. In this second phase of coding, we used the same iterative classification process, revisiting and reclassifying previously categorized graphics as needed. The final subject matter categories are listed and described in Table II using specific examples from the dataset. I then used JMP (Pro 14.2) to compile and analyze the results.

This methodological approach allowed me to carefully categorize both the genre and the subject matter of graphics included in the dataset to identify specific scripto-visual rhetorical strategies that were used during both drug epidemics. As such, the themes that then emerged from my analysis allowed me to determine whether dominant media narratives were being advanced through these choices and to identify those narratives.

RESULTS

This section reports the results of the categories described in Tables I and II. Counts for the categories discussed are given in parentheses. As Fig. 1 shows, bar charts (71) were the most common genre created during the crack epidemic, followed by line graphs (13). Conversely, line graphs (58) were the most common genre created during the opioid epidemic, followed by bar charts (25) and maps (25). All the graphics created during the crack epidemic could be classified using the categories in Table I. In all, 12 of the graphics created during the opioid epidemic were classified as *other* (e.g., dot chart, heat map, histogram).

TABLE II
SUBJECT MATTER CLASSIFICATIONS/THEMES

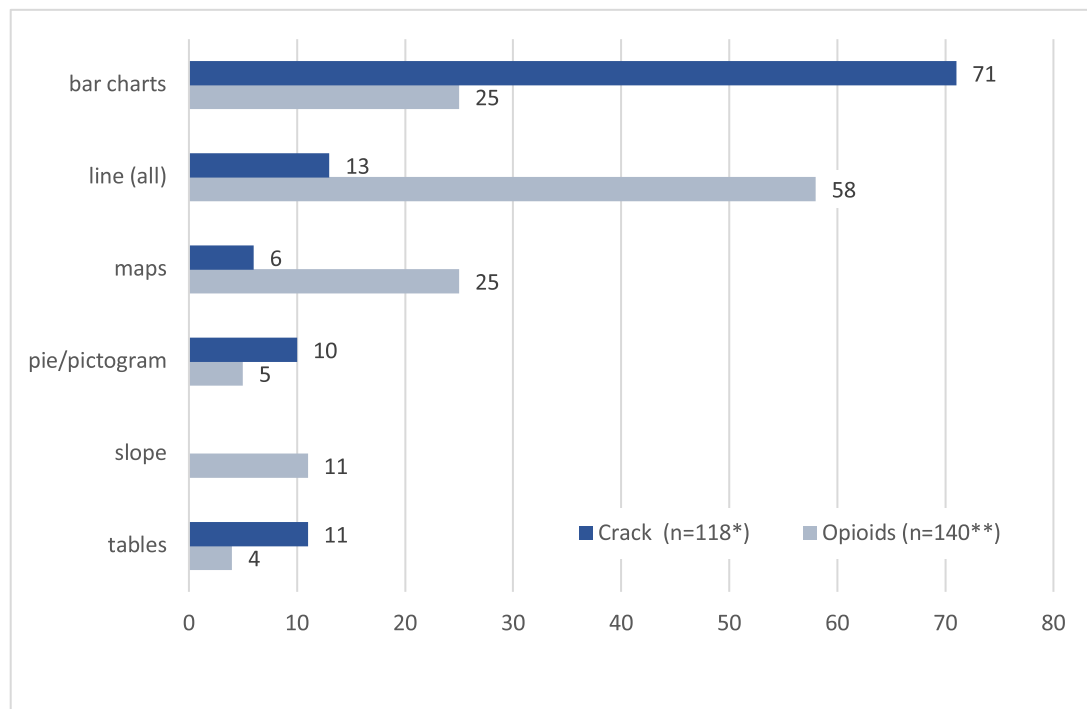
Category	Description and Examples
Crime-related	Includes visuals classified using the following categories from Shachar et al. [8]: • Drugs+crime, e.g., “The Fight Against Drugs: Drug Defendants ...” (LAT1C [62]); • Drugs+trafficking, e.g., “Open Air Drug Markets” (NYT14 [112]); • Investigative, e.g., “Felony Arrests: The Bottom Line” (LAT3A [60]); “Drug Seizures Containing Fentanyl” (NYT82 [113]); • Law+court, e.g., “Sentences for Drug Felonies: Sale of Narcotics” (LAT6A [114]); “Same Crime, Different Punishments” (NYT77 [86]); and • Law+enforcement, e.g., “The Fight Against Drugs: Assets Seized” (LAT1D [62]); “Where Americans Are Sent to Prison Most” (NYT74 [86]).
Health-related	Includes visuals that show quantitative information about the following: • Communicable diseases, e.g., “Sexually Transmitted Diseases ...” (WAPO45A [115]); noncommunicable diseases, e.g., “Heart Disease” (WAPO136A [116]); health conditions, e.g., “Chronic Lower Respiratory Disease” (WAPO136D [116]); and injury, e.g., “Unintentional Injury” (WAPO136C [116]); • Drug abuse treatment, e.g., “More Women in Drug Treatment” (NYT11 [103]); “Drugs to Treat the Effects of Drugs” (WAPO116 [117]); • Drug use, e.g., “Drug Use on the Decline” (NYT20 [118]); “Heavy Drug Use, State by State” (WAPO108 [119]); abuse, e.g., “Cocaine: New Patterns of Abuse: It Is Now the Drug of Choice” (NYT13A [82]); “Heroin Abuse in States ...” (NYT65 [80]); and opioid prescriptions, e.g., “Decline in Opioid Prescriptions” (NYT72 [120]); and the health-related impact/effect of drug use on others, e.g., “Babies Born in NYC Who Tested Positive for Drugs” (NYT17C [83]); “In NM, Drug Addiction Is Being Passed to the Next Generation” (NYT67 [80]).
Mortality	Includes visuals that showed the number of deaths related to drug use, e.g., “Drug-related Deaths” (WAPO34 [121]); caused by drug overdoses, e.g., “Overdose Deaths per 100,000” (NYT62 [80]); and homicides, e.g., “Homicides in D.C. and Baltimore ...” (WAPO51 [122]); “Heroin Deaths Surpass Gun Homicides...” (WAPO127 [123]).
Not Enough Information	Assigned to visuals for which the subject matter could not be determined from the title alone, e.g., “The Corner” (NYT12 [108]) and “A Growing Divide” (NYT73 [86]).

Totally, 11 graphics created during the opioid epidemic were classified as slope graphs (see Fig. 1)—a category not used during the crack epidemic. As the second footnote in Table I explains, slope graphs are a specific category in Emery’s rubric (e.g., see [64]). They differ from line graphs in that they show before and after comparisons between two years. In contrast, line graphs show temporal information; they compare one or more variables over time.

Fig. 2 provides a breakdown of the three types of line graphs found in the dataset: area, traditional, and truncated. Only area line and traditional line graphs were used during the crack epidemic, whereas all three categories were used during the opioid epidemic. Most line graphs created during the opioid epidemic were traditional line graphs (29), followed by truncated (21), and then area line graphs, as Fig. 2 shows.

In terms of subject matter categories (see Table II) and as Fig. 3 shows, the titles of most graphics

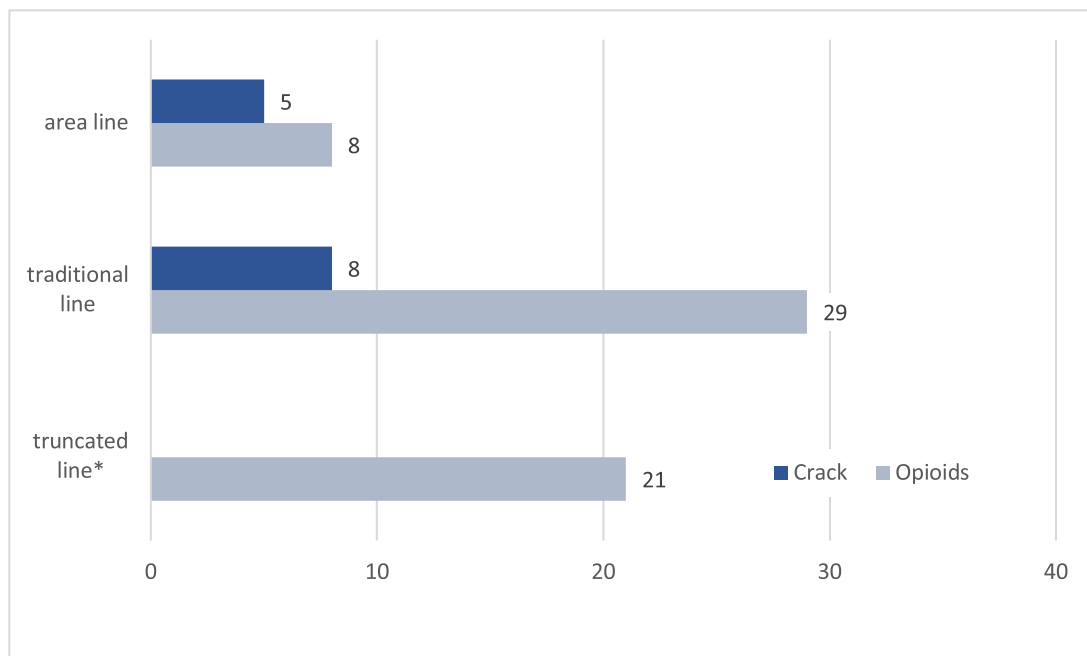
created during the crack epidemic that could be classified were categorized as *crime-related only* (44). In contrast, the titles of most graphics created during the opioid epidemic that could be classified were categorized as *mortality only* (57). The number of graphics categorized as *health-related only* was comparable (32 for crack and 40 for opioids), whereas the number categorized as *not enough info* was nearly identical. In all, 14 were classified as *crime-related; mortality* during the crack epidemic with only 2 falling into this classification for the opioid epidemic. In total, 11 graphics were also classified into combination categories like *crime-related; health-related* (e.g., WAPO35A [70]; WAPO171 [71]) and *health-related; mortality* (WAPO120B [72]), meaning that the subject matter represented in the graphic could be classified as both categories. These categories are not shown in Fig. 3 because there were fewer than five graphics for each of these combination classifications. *Other* is also not shown or described in Table II because only six graphics from both epidemics fell into this



*Seven (7) graphics were classified as histograms; no graphics were classified as slope graphs.

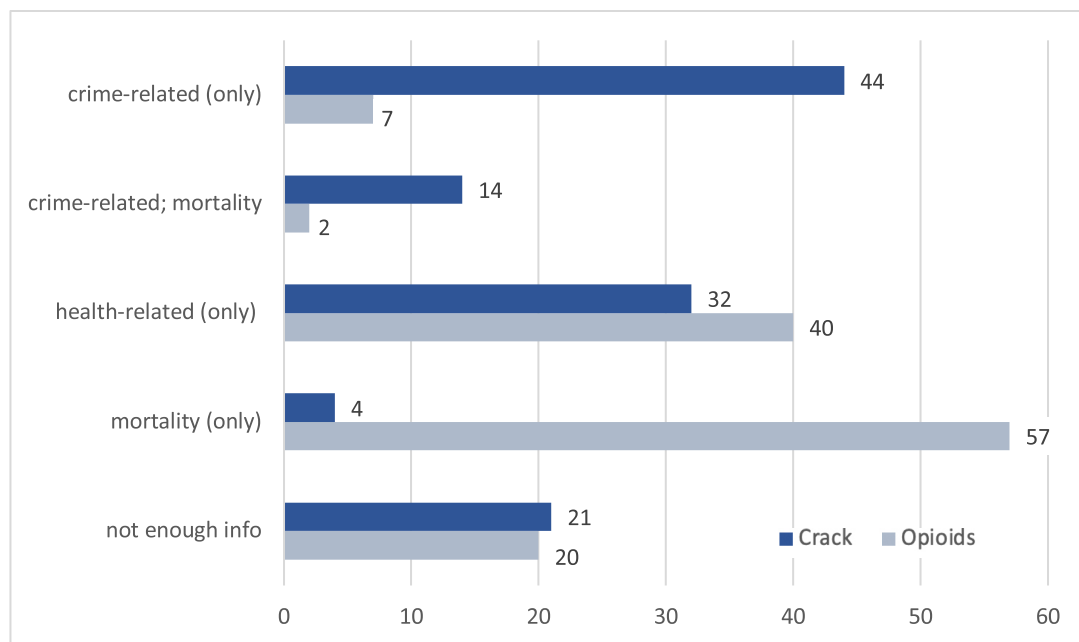
**Twelve (12) graphics were classified as *other* as described in the Results.

Fig. 1. Distribution of graphics by genre: Both epidemics.



*The truncated line category for opioids includes five (5) truncated area line graphs.

Fig. 2. Distribution of line graphs: Both epidemics.



Six (6) graphics from both epidemics were classified as other, and eleven (11) were classified as combination categories (e.g., crime-related; health-related). These categories are described in the Results.

Fig. 3. Number of graphics by subject matter categories: Both epidemics.

category. *Other* was assigned to visuals where the subject matter could be determined but there were no other graphics that could be assigned to the category such as “D.C. Child Welfare Social Workers ...” (WAO48 [73]) and “Thousands More in Foster Care” (NYT79 [74]).

Figs. 4 and 5 provide a breakdown of the most common genres—bar charts, line graphs, and maps (see Fig. 1)—by subject matter for both epidemics, aligning with the results shown in Figs. 1 and 3. More specifically, as Fig. 4 shows, bar charts during the crack epidemic showed *crime-related* (30) information most commonly. Conversely, line graphs and maps were commonly used to show *mortality* (28 and 13, respectively) during the opioid epidemic (see Fig. 5).

ANALYSIS: THE SCRIPTO-VISUAL RHETORIC OF THE DATASET

As the previous section demonstrates, the results of the subject matter portion of this study align closely with Shachar et al.’s [8] findings. More specifically, crime-related quantitative information was predominantly visualized during the crack epidemic in the dataset created for this study.

Conversely, mortality, which is a specific public health outcome, was predominantly visualized during the opioid epidemic in the dataset, supporting the focus on health and public health-related themes that Shachar et al. [8] found in coverage of the opioid epidemic. Yet, as previously stated, data visualizations construct meaning through both visual and linguistic/textual semiotic modes—that is, through the subject matter of the graphic and the visual relationships conveyed through the genre. Thus, the results discussed in the previous section also describe trends in the scripto-visual rhetoric of the graphics included in this study. In this section, I draw from research in sociology on media coverage during outbreaks to argue that these scripto-visual rhetorical choices escalated the threat posed during *both* drug epidemics while also reinforcing dominant media narratives that positioned crack addiction as a problem needing an immediate law enforcement response but opioid addiction as a long-term public health issue.

Research in sociology has argued that media coverage during outbreaks follows a reporting trajectory that shifts from alarming to reassuring as audiences become more frightened [29], [75]. More specifically, the alarmist messaging phase is

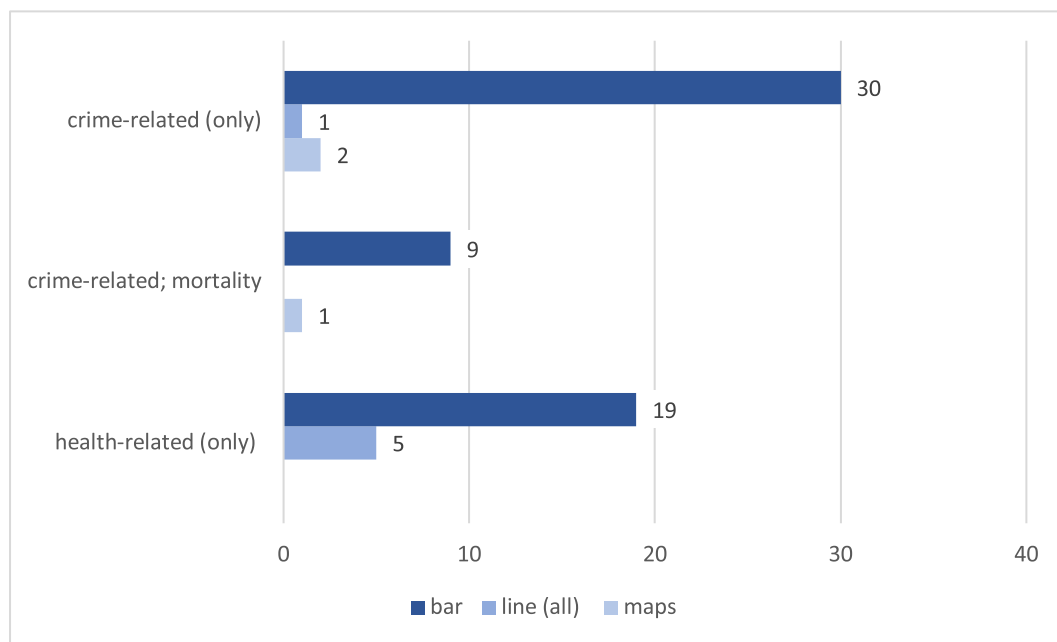


Fig. 4. Crack epidemic: Breakdown by genre and subject matter theme.

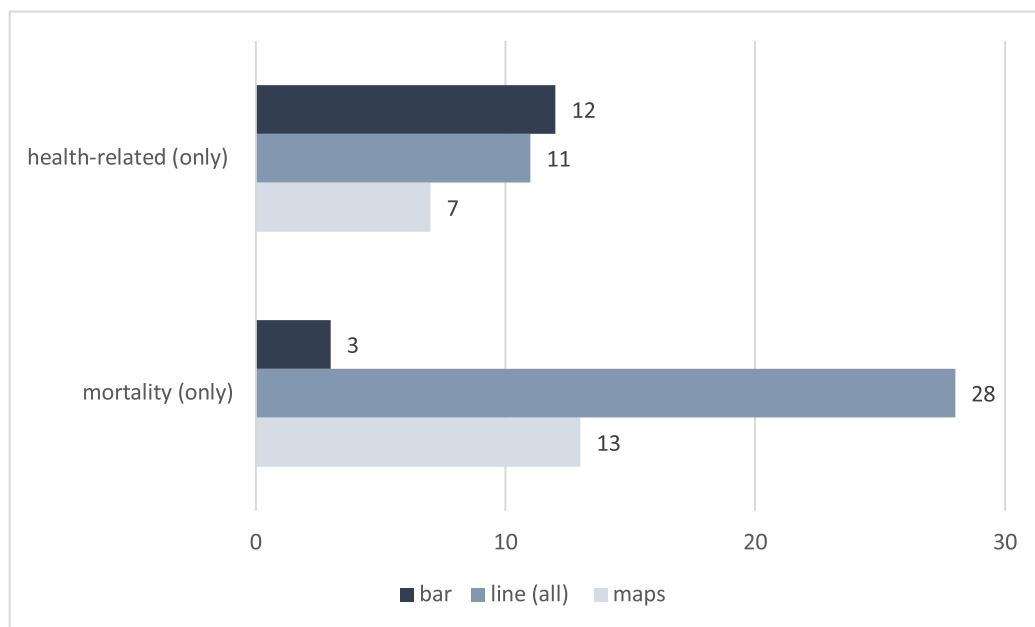


Fig. 5. Opioid epidemic: Breakdown by genre and subject matter theme.

characterized by “fearful claims making” [29, p. 480] through frequent news reports emphasizing “incredible numbers” [p. 483–484] such as a potentially catastrophic number of deaths. This emphasis on the number of people affected characterizes the more expansive definition of epidemic commonly used today, which “can be said to exist whenever the number of cases exceeds that

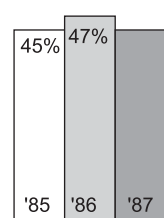
which is considered normal or expected based upon past experience for a given population” [30, p. 32]. Indeed, both the crack and opioid situations have routinely been described as epidemics. Furthermore, the data visualizations published during both situations escalated the perceived threat by emphasizing incredible numbers. To illustrate, NYT15 [76] (see Fig. 6),

Drugs: Murder's Common Denominator

Figures for the District of Columbia

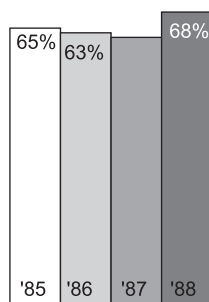
Assailants

Percent of people arrested for murder who tested positive for drugs. About half of all people arrested where tested. Figures for 1988 were not available.



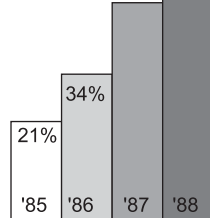
Victims

Murder victims testing positive for drugs or alcohol.



Motives

Percent of murders (where motive was known) that were motivated by drugs and drug activities.



Source: District of Columbia Office of Criminal Justice

The New York Times/March 28, 1989

Fig. 6. NYT15 was published in *The New York Times* on March 28, 1989, in an article entitled “Capital Offers a Ripe Market to Drug Dealers” [76, p. A16]. The values for 1987 could not be included because this information could not be determined from the original image. From *The New York Times*. © 1989 The New York Times Company. All rights reserved. Used under license. <https://www.nytimes.com/>

published during the crack epidemic, and LAT60 [77] (not shown), published during the opioid epidemic, both show increases in the variables depicted. However, each employed different scripto-visual reporting strategies that were used to construct very different media narratives.

More specifically, bar charts showing crime-related information were published more frequently during the crack epidemic, whereas line graphs showing mortality (like NYT86 [78]); not shown) were published more frequently during the opioid epidemic. Rhetorically, bar charts emphasize comparisons, as NYT15 demonstrates (see Fig. 6), which shows “Drugs: Murder’s Common Denominator” and percentage increases for 1985, 1986, and 1987 for “Assailants,” “Victims,” and “Motives” [76]. (The last two categories also include percentages for 1988.) This graphic does show a trend over time for these variables. However, its core visual rhetorical function is to convey changes *between* each of the years being compared. As we can see, little change has occurred in the “[p]ercent of people arrested for murder who tested positive for drugs” (“Assailants”), and “[m]urder victims testing positive for drugs or alcohol” (“Victims”). At the same time, a steady and dramatic increase can be seen between the years shown in

“Motives”—that is, the “[p]ercent of murders ...that were motivated by drugs and drug activities.” The scripto-visual rhetoric of this specific portion of NYT15, then, is that the motives for this form of violent crime (i.e., murder) have surged each year over the past three years.

Furthermore, because this graphic shows crime-related information, it frames the issue as a criminal justice problem, which requires a law-and-order solution. The visual genre used (i.e., bar chart) thus heightens the urgency of addressing the increase in murders that can be attributed to “drugs and drug activities” (i.e., crack in this context). More specifically, this graphic argues that this problem began in 1985 and has only become worse in the three years since (growing from 21% in 1985 to 80% in 1988). Indeed, NYT15, published in an article entitled “Capital Offers a Ripe Market to Drug Dealers” [76], describes the increasing severity of the crack epidemic in Washington, DC, along with the criminal justice solutions being enacted to control the situation.

During the crack epidemic, bar graphs were also used to show longer trends over time, as WAPO29 [79] (see Fig. 7) demonstrates, documenting “drug arrests in DC” from 1977 to 1987. The creators of this graphic could have used a line graph to convey this information. However, like NYT15 (see Fig. 6), this graphic visually calls attention to the increase between each of the years shown, emphasizing the consistent and steady increase in “drug arrests.”

In contrast, a line graph would have smoothed over any fluctuations between the years shown, emphasizing the overall dominant trend, as LAT60 [77] (not shown) illustrates. More specifically, LAT60, a traditional line graph published during the opioid epidemic, shows growing numbers of “opioid-related hospital emergency department visits” from 2007 to 2015. However, a closer examination of the line reveals that there is not always an increase between years. Using a bar chart would have called attention to these variations, whereas the line reinforces this graphic’s overall “crisis trend” argument.

Bar graphs are more rhetorically effective in emphasizing changes between years because the bars emphasize comparisons between variables. Indeed, the line graphs published during the opioid epidemic in the dataset showing mortality-related information (see Fig. 5)—often for a decade or longer—served to position opioid addiction as a long-term public health problem. As with the crack

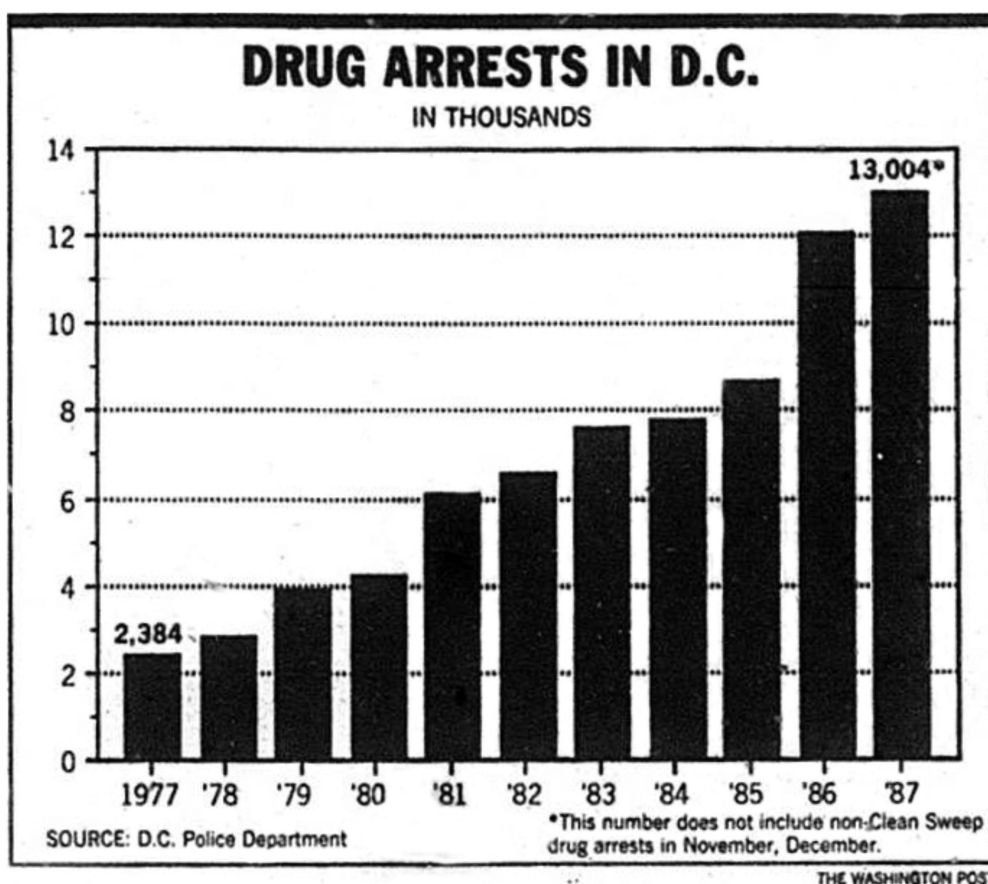


Fig. 7. WAPO29 was published in *The Washington Post* on January 26, 1988, in an article entitled “Operation Clean Sweep’s Future Uncertain” [79, p. A14]. From *The Washington Post*. © 1988 *The Washington Post*. All rights reserved. Used under license. <https://www.washingtonpost.com/>

epidemic, these graphics visually escalated the perceived threat of the drug epidemic by constructing a visual argument that the situation urgently needed to be addressed. Yet, because the subject matter is mortality, public health-oriented solutions are the appropriate response.

During the opioid epidemic, the visual narrative emphasizing public health solutions was further bolstered by using two graphical genres not seen during the crack epidemic as previously identified: truncated line graphs (see Fig. 2 and Table I) and slope graphs (see Fig. 1 and Table I). More specifically, the former uses the visual conventions of a traditional line graph but employs a shorter line as well as labels for only the first and last data points on the *x*-axis, giving the line a shortened or truncated appearance. Like traditional line graphs, these design choices downplay changes between years, but they also visually exaggerate the dominant trend, erasing inconsistencies (if any) between the years shown. For instance, NYT64

([80]; not shown) compares overdose deaths from opioids in rural areas to those in large metropolitan areas from 2002 to 2014, using a shorter line than would be seen in a traditional line graph and labeling only these two years. The result is a far more dramatic curve in the line, amplifying the overall upward trend.

The second category—slope graphs—also uses a line. However, these visuals are not line graphs (see Table I) because they emphasize before/after comparisons [64], as previously explained, using a single, straight line to compare two points in time rather than to show a trend. WAPO99A [81] (not shown), for instance, depicts mortality rates for three different races for 1999 and 2013 only, and connects these two years with a short, straight line. This visual strategy does more than just mask variations in the directionality of the line; it erases them. More to the point, it condenses the trend into two binary categories—in this case, the numbers for 1999 and for 2013, respectively. As such, these

Cocaine: New Patterns of Abuse

Figures for people entering Phoenix House for treatment each year. Crack appeared on New York City streets in mid-1985.

It Is Now the Drug of Choice . . .

Number of people seeking drug treatment who reported using each drug.

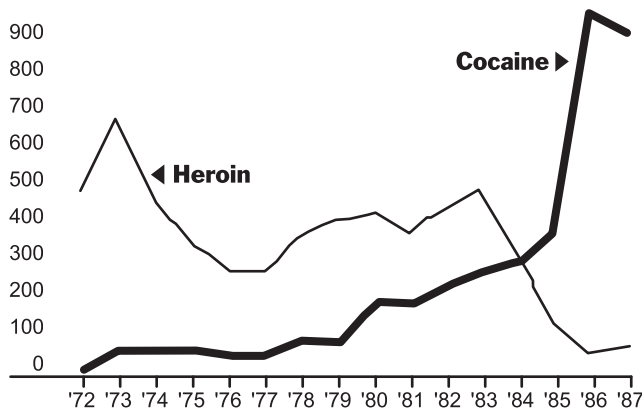


Fig. 8. NYT13A was published in *The New York Times* on June 23, 1988, in an article entitled “Addiction’s Hidden Toll: Poor Families in Turmoil” [82, p. 28]. From *The New York Times*. © 1988 The New York Times Company. All rights reserved. Used under license. <https://www.nytimes.com/>

scripto-visual rhetorical strategies—using truncated line and slope graphs to show mortality data in this dataset—served to amplify the perceived threat posed by the opioid epidemic. And again, because the subject matter shown is mortality data, a public health response is the appropriate solution.

In fact, opioid addiction is a long-term public health problem (see [45]), which was not the case specifically for crack when news coverage about this drug began to escalate. However, as line graphs like NYT8A (“Trends in Cocaine Use” [56]; not shown) and NYT13A (“Cocaine: New Patterns of Abuse” [82]; see Fig. 8) demonstrate, cocaine abuse was also a long-term health-related problem during the crack epidemic. In addition, these visuals as well as others like NYT17A entitled “Emergency Room Visits Related to Crack” ([83]; not shown; see also [84], [85]) demonstrate that health-related data were visualized during the crack epidemic (see Fig. 3). These data could have been used to construct a very different overall media narrative. In addition, crime-related data, too, were published during the opioid epidemic as Fig. 3 demonstrates (e.g., see [72], [86], [87]). Yet quantitative health-related information was not visually emphasized during the crack epidemic, and quantitative crime-related information was not

visually emphasized during the opioid epidemic in the datasets created for this study.

DISCUSSION: ENACTING A SOCIAL JUSTICE ETHIC IN THE CONSTRUCTION OF DATA VISUALIZATIONS

It is important to acknowledge that the crack epidemic occurred within a very different social, cultural, and political context than we have experienced with the current opioid situation. More specifically, crack emerged when the US was deeply entrenched in the war on drugs, which may have eliminated the possibility that this drug epidemic would have been positioned differently during this period. Indeed, Shachar et al. [8] confirmed that none of the articles in their study frames the crack epidemic as a public health issue.

In the 30 years separating the two drug epidemics discussed in this article, the capabilities of data design software, too, have grown exponentially, allowing for dramatically more visually sophisticated representations that can be seen in many of the graphics published during the opioid epidemic. These tools, too, have facilitated the emergence of novel forms (e.g., slope graphs) and techniques (e.g., truncated line graphs), as Table I shows. Indeed, all of the graphics published during the crack epidemic included in this study are print based, whereas all of those published during the opioid epidemic are digital. Thus, during the crack epidemic, creators may have also had to attend to space limitations in their graphical design choices, a constraint far less likely to affect designers today.

A study that explored the genre used in print-based graphics from 1985 to 1994 found that bar charts and line graphs were the most published forms in the newspapers surveyed [88], a conclusion that aligns with the results of this study. That study, conducted by Zacks et al., included the same newspapers used in the present study, as well as *The Wall Street Journal* [88]. Indeed, prior to 1980, *The New York Times* routinely published line graphs only in the business section, whereas the invention of the personal computer enabled the widespread inclusion of these visuals in other sections and facilitated the emergence of the design style used in newspapers that can still be seen today [59]. Furthermore, data visualizations today continue to rely on core visual metaphors (e.g., lines, bars, shading), design conventions that remain largely unchanged since they were developed in the 19th century [89], and most of the graphics in this study could be classified using

well-established visual forms (e.g., line graphs, bar charts, maps).

As an additional limitation of this study, including only the titles in the linguistic/textual analysis portion may have prevented a more in-depth and nuanced exploration of the linguistic/textual rhetoric used in these visuals, and I initially sought to analyze this aspect. Yet it quickly became apparent that this content was widely inconsistent. Not all graphics had captions, the length of captions varied considerably, and many graphics were not discussed in the body of the article in which they were published. Thus, the title proved to be the only linguistic feature that could be consistently compared. Furthermore, the title is arguably the central component of a data visualization's overall linguistic message because it conveys the main gist of the subject matter to readers.

That said, the recent and dramatic shift toward “favor[ing] treatment over incarceration,” as Hart [6, para. 4] characterized it, seen in coverage of the opioid epidemic, might be attributed, in part, to changing attitudes toward some drugs [90], drug-related offenses [91], and addiction (e.g., see [92], [93]). Case and Deaton, whose landmark study brought national media attention to the opioid epidemic, too, have been strongly criticized for turning unnecessary attention toward the most privileged racial group in the US [94]. In response, they stated in an interview with *The Washington Post* that their work “was very much just the facts,” [94, para. 10], referencing a line graph published in their original report [1]. This interaction illustrates Walton et al.'s [35] point that inequalities are often hidden through the rhetoric of neutrality frequently ascribed to technical communication artifacts while also demonstrating how positionality, privilege, and power get enacted in ways that promote these inequalities in particular communicative contexts [34].

More specifically, by marshaling this graphic as irrefutable evidence (i.e., “just the facts”), Case and Deaton invoke their scientific authority [95] to shut down criticism of their work while also simultaneously reinforcing the positivist assumptions that still often undergird how nonexpert audiences perceive these visuals—that is, as neutral, self-evident depictions of reality [96]. They use their position (i.e., their identity as well-respected economists) and the privilege that this identity confers (i.e., scientific credibility and authority), combined with the power of these

graphics [97], to position their data visualization decisions as neutral.

When creators of these graphics (e.g., subject matter experts, technical communicators, visual journalists) make choices about what datasets to visualize—as scholarship in TPC has pointed out [98], [99]—and how to visualize them (i.e., their scripto-visual rhetorical choices), the underlying values that inform those decisions are endorsed and promoted. When these data visualizations are subsequently widely circulated to broad, public audiences through established and well-respected media outlets like those included in this study, these arguments become normalized. Thus, choosing to position people addicted to crack as criminals but people addicted to opioids as patients also becomes a profoundly ethical decision, demonstrating that a social justice ethic needs to be integrated into data design work.

Toward this end, I draw from Jones et al.'s positionality, privilege, and power framework [34], [35] to propose the heuristic outlined below. These questions can be used as intentional prompts when visualizing data or as an analytical tool to tease out the effect of a creator's specific scripto-visual rhetorical choices.

- What identity markers (such as “race, ethnicity, gender, sexuality, ableness, religion, class” [34, p. 220]) are represented in the dataset selected?
- What identity markers are included in the specific variables that will be visualized? What relationships (e.g., comparative, spatial, and temporal) will be shown between and among these variables, and how do these relationships position the subject matter of the dataset?
- How might these scripto-visual rhetorical choices—the subject matter and genre chosen by the creator—position, privilege, and empower certain groups and/or perspectives?
- How might these scripto-visual rhetorical choices disenfranchise or marginalize vulnerable groups or perspectives? What changes can be made that might empower these groups?

To illustrate how this heuristic might be applied analytically, next I evaluate NYT13A [82] (see Fig. 8), a line graph published during the crack epidemic and discussed at the end of the previous section. This graphic, as well as others [84], [85], as previously stated, shows that health-related data were, in fact, visualized during the crack epidemic, confirming that the focus on crime-related information that I observed in the dataset that I

created was a rhetorical decision. I analyze this visual to demonstrate that enacting a social justice ethic requires identifying the ways that scripto-visual rhetorical decisions may be privileging specific identities, groups, or perspectives.

To begin, NYT13A (Fig. 8) shows that the number of people seeking treatment for heroin decreased dramatically from 1972 to 1987 but those seeking treatment for cocaine abuse surged. The identity markers represented are heroin and cocaine users, respectively. This traditional line graph is used to construct two temporal relationships: trends in users of each of these drugs seeking treatment over a 15-year period. Rhetorically, the scripto-visual rhetoric of this graphic positions crack as a rapidly escalating threat, demonstrated by the “incredible number” [29, p. 483] of cocaine users compared to heroin users seeking treatment.

Although the visual construction of this argument may seem straightforward thus far, applying the last two questions from the heuristic I have proposed is revealing. More specifically, the profound dip in heroin users positions the need for treatment for this drug as dramatically declining, whereas the inverse is true for cocaine users. Both groups of illegal drug users represented in this graphic are “precarious publics” [100] and thus subject to disenfranchisement and marginalization. Yet, arguably, the graphic’s scripto-visual rhetoric “privileges” cocaine users. It was published in a story about the effect of the crack epidemic on families and could have been used to “empower” or otherwise call attention to the increased need for treatment for cocaine users *if* addiction had been framed as a public health issue—perhaps by arguing that more resources were needed. Such framing, however, might also have simultaneously disenfranchised people addicted to heroin.

Depending on the context, then, the scripto-visual rhetoric of this graphic may not necessarily be advocating inequitable treatment for people experiencing drug addiction. However, the positionality of people addicted to either of these drugs is potentially equally precarious. Applying the heuristic analytically calls attention to these aspects of the graphic’s scripto-visual rhetoric so that the appropriateness of these decisions can be determined, reinforcing the ethical decision-making practice of determining how best to visually convey this information about people needing treatment for illegal drug use. If reducing support for people seeking treatment for heroin is disenfranchising,

this visual information could be removed, focusing the argument only on the trend involving cocaine use.

CONCLUSIONS

A social justice perspective calls attention to communication strategies that can empower marginalized groups [101], as well as decisions that promote inequality [34], which has been the emphasis of this study. By foregrounding the ways that the scripto-visual rhetoric of data visualizations may be positioning, privileging, and empowering certain groups and thus reinforcing inequities, the heuristic that I propose can inform technical communication work in three areas—in our scholarship, in our teaching, and in our role as user advocates.

To illustrate, technical communication scholars who collect quantitative information on human subjects—such as those working in health and medicine—can apply the heuristic when they make decisions about which variables to visualize from the datasets that they construct. In this context, the heuristic’s questions can urge researchers to consider the identity markers they may be privileging in choosing to visualize certain variables over others, as well as how these choices might empower or marginalize certain groups or perspectives.

The heuristic can also be used in pedagogical practice to prompt students to critique the data visualization decisions that creators of these graphics have already made (as I have done in the previous section). Analytically applying the heuristic can foster in-depth class discussions about the rhetorical power of these visuals while also encouraging students to consider how positionality, privilege, and power might be enacted in the data design decisions that they make in their own work. Indeed, many technical communication textbooks provide instruction in which graphical form to use to create a particular rhetorical effect (e.g., use a line graph to show a trend over time). However, the ways that these decisions might advance power relationships do not tend to be addressed. Using the heuristic with students also iterates that data visualizations are rhetorical—that is, that these graphics advance visual arguments about how data should be interpreted.

Finally, as technical communication experts, one of our key roles is to advocate for users. Indeed, many of us may be working (or have previously worked)

as practitioners or we may be in situations where we can influence how subject matter experts convey visual quantitative information to nonexpert audiences. In these contexts, the heuristic I propose can be used when advising creators of these graphics by prompting them to consider the effects of their scripto-visual rhetorical choices more fully.

In closing, the heuristic that I offer can be used to incorporate a social justice ethic into data design work by calling attention to the ways that creators' scripto-visual rhetorical decisions can position, privilege, or empower certain groups or

perspectives. As such, the questions in the heuristic also simultaneously push back against the neutral positionality frequently ascribed to these visuals, lending insight into the ways that these graphics might be disenfranchising or marginalizing (either advertently or inadvertently) particular groups. Through such efforts, I suggest, we can continue moving toward more equitable communication practices in TPC in our efforts to achieve "active equality" [102].

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REFERENCES

- [1] A. Case and A. Deaton, "Rising morbidity and mortality in midlife among white non-Hispanic Americans in the 21st century," *Proc. Nat. Acad. Sci. USA*, vol. 112, no. 49, pp. 15078–15083, 2015. [Online]. Available: <http://www.pnas.org/content/112/49/15078.full>
- [2] G. Kolata, "Deaths rates rising for middle-aged white Americans, study finds," *The New York Times*, Nov. 2, 2015. [Online]. Available: <https://www.nytimes.com/2015/11/03/health/death-rates-rising-for-middle-aged-white-americans-study-finds.html>
- [3] Office of National Drug Control Policy. (n.d.). President's commission on opioids. [Online]. Available: <https://trumpwhitehouse.archives.gov/ondcp/the-administrations-approach/presidents-commission-opioids>
- [4] K. Demirjian, "Congress set to pass bill to combat opioid abuse," *The Washington Post*, Jul. 10, 2016. [Online]. Available: https://www.washingtonpost.com/politics/congress-set-to-pass-bill-to-combat-opioid-abuse/2016/07/10/a5c6b52a-46cb-11e6-bdb9-701687974517_story.html?utm_term=.2d8eab52e9ce
- [5] D. Merica, "Trump declares opioid epidemic a national public health emergency," *CNN*, Oct. 26, 2017. [Online]. Available: <http://www.cnn.com/2017/10/26/politics/donald-trump-opioid-epidemic/index.html>
- [6] C. L. Hart, "The real opioid emergency," *The New York Times*, Aug. 18, 2017. [Online]. Available: <https://www.nytimes.com/2017/08/18/opinion/sunday/opioids-drugs-race-treatment.html>
- [7] E. Yankah, "There was no wave of compassion when addicts were hooked on crack," *PBS News Hour*, Mar. 29, 2016. [Online]. Available: <https://www.pbs.org/newshour/show/there-was-no-wave-of-compassion-when-addicts-were-hooked-on-crack>
- [8] C. Shachar, T. Wise, G. Katznelson, and A. L. Campbell, "Criminal justice or public health: A comparison of the representation of the crack cocaine and opioid epidemics in the media," *J. Health Politics Policy Law*, vol. 45, no. 2, pp. 211–239, 2020, doi: [10.1215/03616878-8004862](https://doi.org/10.1215/03616878-8004862).
- [9] R. M. Entman, "Framing: Toward clarification of a fractured paradigm," *J. Commun.*, vol. 43, no. 4, pp. 51–58, 1993.
- [10] D. A. Scheufele, "Framing as a theory of media effects," *J. Commun.*, vol. 49, no. 1, pp. 103–122, 1999.
- [11] K. Beckett, "Setting the public agenda: 'Street crime' and drug use in American politics," *Social Problems*, vol. 41, no. 3, pp. 425–447, 1994.
- [12] W. A. Gamson and A. Modigliani, "Media discourse and public opinion on nuclear power: A constructionist approach," *Amer. J. Sociol.*, vol. 95, no. 1, pp. 1–37, 1989.
- [13] M. Alexander, *The New Jim Crow: Mass Incarceration in the Age of Colorblindness*. New York, NY, USA: New Press, 2012.
- [14] L. D. Bobo and V. Thompson, "Unfair by design: The war on drugs, race, and the legitimacy of the criminal justice system," *Social Res., Int. Quart.*, vol. 73, no. 2, pp. 445–472, 2006.
- [15] J. Fellner, "Race, drugs, and law enforcement," *Stanford Law Policy Rev.*, vol. 20, no. 2, pp. 257–291, 2009.
- [16] L. Alexandrescu, "Mephedrone, assassin of youth: The rhetoric of fear in contemporary drug scares," *Crime Media Culture*, vol. 60, no. 1, pp. 23–37, 2014, doi: [10.1177/1741659013511975](https://doi.org/10.1177/1741659013511975).
- [17] M. M. Orsini, "Frame analysis of drug narratives in network news coverage," *Contemp. Drug Problems*, vol. 44, no. 3, pp. 189–211, 2017.
- [18] D. Russell, N. J. Spence, and K. M. Thames, "It's so scary how common this is now': Frames in media coverage of the opioid epidemic by Ohio newspapers and themes in Facebook user reactions," *Inf. Commun. Soc.*, vol. 22, no. 5, pp. 702–708, 2019, doi: [10.1080/1369118X.2019.1566393](https://doi.org/10.1080/1369118X.2019.1566393).
- [19] N. Swalve and R. DeFoster, "Framing the danger of designer drugs: Mass media, bath salts, and the 'Miami Zombie Attack,'" *Contemp. Drug Problems*, vol. 43, no. 2, pp. 103–121, 2016, doi: [10.1177/0091450916642515](https://doi.org/10.1177/0091450916642515).
- [20] R. R. Weidner, "Methamphetamine in three small Midwestern cities: Evidence of a moral panic," *J. Psychoactive Drugs*, vol. 41, no. 3, pp. 227–239, 2009.

- [21] E. Willis and C. Painter, "The needle and the damage done: Framing the heroin epidemic in the Cincinnati Enquirer," *Health Commun.*, vol. 34, no. 6, pp. 661–671, 2018, doi: [10.1080/10410236.2018.1431023](https://doi.org/10.1080/10410236.2018.1431023).
- [22] H. Hansen, "Assisted technologies of social reproduction: Pharmaceutical prosthesis for gender, race, and class in the white opioid 'crisis,'" *Contemp. Drug Problems*, vol. 44, no. 4, pp. 321–338, 2017, doi: [10.1177/0091450917739391](https://doi.org/10.1177/0091450917739391).
- [23] J. Netherland and H. B. Hansen, "The war on drugs that wasn't: Wasted whiteness, 'Dirty doctors,' and race in media coverage of prescription opioid misuse," *Culture Med. Psychiatry*, vol. 40, no. 4, pp. 664–686, 2016, doi: [10.1007/s11013-016-9496-5](https://doi.org/10.1007/s11013-016-9496-5).
- [24] D. Jernigan and L. Dorfman, "Visualizing America's drug problems: An ethnographic content analysis of illegal drug stories on the nightly news," *Contemp. Drug Problems*, vol. 23, no. 2, pp. 169–196, 1996.
- [25] J. D. Orcutt and J. B. Turner, "Shocking numbers and graphic accounts: Quantified images of drug problems in the print media," *Social Problems*, vol. 40, no. 2, pp. 190–206, 1993.
- [26] L. Cooke, "Information acceleration and visual trends in print, television, and web news sources," *Tech. Commun. Quart.*, vol. 12, no. 2, pp. 155–182, 2003.
- [27] H. Dawkins, "Ecology, images, and scripto-visual rhetoric," in *Ecosee: Image, Rhetoric, Nature*, S. I. Dobrin and S. Morey, Eds. Albany, NY, USA: State Univ. New York Press, 2009, pp. 79–94.
- [28] D. Ross, "The role of ethics, culture, and artistry in scientific illustration," *Tech. Commun. Quart.*, vol. 26, no. 2, pp. 145–172, 2017, doi: [10.1080/10572252.2017.1287376](https://doi.org/10.1080/10572252.2017.1287376).
- [29] S. Ungar, "Global bird flu communication: Hot crisis and media reassurance," *Sci. Commun.*, vol. 29, no. 4, pp. 472–497, 2008, doi: [10.1177/1075547008316219](https://doi.org/10.1177/1075547008316219).
- [30] D. D. Chitwood, S. Murphy, and M. Rosenbaum, "Reflections on the meaning of drug epidemics," *J. Drug Issues*, vol. 30, no. 1, pp. 29–39, 2009.
- [31] N. Allen, "Ethics and visual rhetorics: Seeing's not believing anymore," *Tech. Commun. Quart.*, vol. 5, no. 1, pp. 87–105, 1996.
- [32] S. Dragga and D. Voss, "Cruel pies: The inhumanity of technical illustrations," *Tech. Commun.*, vol. 48, no. 3, pp. 265–274, 2001.
- [33] N. Amare and A. Manning, "The language of visuals: Text + graphics = visual rhetoric tutorial," *IEEE Trans. Prof. Commun.*, vol. 50, no. 1, pp. 57–70, Mar. 2007.
- [34] N. N. Jones, K. R. Moore, and R. Walton, "Disrupting the past to disrupt the future: An antenarrative of technical communication," *Tech. Commun. Quart.*, vol. 25, no. 4, pp. 211–229, 2016, doi: [10.1080/10572252.2016.1224655](https://doi.org/10.1080/10572252.2016.1224655).
- [35] R. Walton, K. Moore, and N. Jones, *Technical Communication After the Social Justice Turn: Building Coalitions for Action*. New York, NY, USA: Routledge, 2019.
- [36] E. Goffman, *Frame Analysis: An Essay on the Organization of Experience*. Cambridge, MA, USA: Harvard Univ. Press, 1974.
- [37] R. Coleman, "Framing the pictures in our heads," in *Doing News Framing Analysis: Empirical and Theoretical Perspectives*, P. D'Angelo and J. Kuypers, Eds. New York, NY, USA: Routledge, 2010, pp. 233–261.
- [38] D. L. Altheide, "The news media, the problem frame, and the production of fear," *Sociol. Quart.*, vol. 38, no. 4, pp. 647–668, 1997, doi: [10.1111/j.1533-8525.1997.tb00758.x](https://doi.org/10.1111/j.1533-8525.1997.tb00758.x).
- [39] D. L. Altheide and C. J. Schneider, *Qualitative Media Analysis*, 2nd ed. Thousand Oaks, CA, USA: Sage, 2013.
- [40] K. Beckett, "Media depictions of drug abuse: The impact of official sources," *Res. Pol. Soc.*, vol. 7, pp. 161–182, 1995.
- [41] S. Cohen, *Folk Devils and Moral Panics: The Creation of the Mods and the Rockers*, 3rd ed. New York, NY, USA: Routledge, 2002.
- [42] S. L. Muzzatti, "Bits of falling sky and global pandemics: Moral panic and severe acute respiratory syndrome (SARS)," *Illness, Crisis Loss*, vol. 13, no. 2, pp. 117–128, 2005.
- [43] T. Chiricos, "Moral panic as ideology: Drugs, violence and punishment in America (1996)," in *Critical Readings: Moral Panics and the Media*, C. Critcher, Ed. Maidenhead, UK: Open Univ. Press, 2006, pp. 103–123.
- [44] P. M. V. Martin and E. Martin-Granel, "2,500-year evolution of the term epidemic," *Emerg. Infect. Dis.*, vol. 12, no. 6, pp. 976–980, 2006, doi: [10.3201/eid1206.051263](https://doi.org/10.3201/eid1206.051263).
- [45] Centers for Disease Control and Prevention. (2018 Dec. 19). Understanding the epidemic. [Online]. Available: <https://www.cdc.gov/drugoverdose/epidemic/index.html>
- [46] E. E. McGinty, A. Kennedy-Hendricks, J. Baller, J. Niederdeppe, S. Gollust, and C. L. Barry, "Criminal activity or treatable health condition? News media framing of opioid analgesic abuse in the United States, 1998–2012," *Psychiatric Serv.*, vol. 67, no. 4, pp. 405–411, 2016, doi: [10.1176/appi.ps.201500065](https://doi.org/10.1176/appi.ps.201500065).
- [47] D. F. Musto, *The American Disease: Origins of Narcotic Control*, 3rd ed. New York, NY, USA: Oxford Univ. Press, 1999.
- [48] N. Murakawa, "Toothless: The methamphetamine 'epidemic,' 'meth mouth,' and the racial construction of drug scares," *Du Bois Rev., Social Sci. Res. Race*, vol. 8, no. 1, pp. 219–228, 2011, doi: [10.1017/S1742058X11000208](https://doi.org/10.1017/S1742058X11000208).
- [49] C. Reinerman and H. G. Levine, "Crack in context: Politics and media in the making of a drug scare," *Contemp. Drug Problems*, vol. 16, no. 4, pp. 535–578, 1989.
- [50] J. P. Hoffmann, "The historical shift in the perception of opiates: From medicine to social menace," *J. Psychoactive Drugs*, vol. 22, pp. 53–62, 1990.
- [51] M. Szalavitz, "What the media gets wrong about opioids," *Columbia Journalism Rev.*, Aug. 15, 2018. [Online]. Available: https://cjr.org/covering_the_health_care_fight/what-the-media-gets-wrong-about-opioids.php

- [52] P. Messaris and L. Abraham, "The role of images in framing news stories," in *Framing Public Life: Perspectives on Media and Our Understanding of the Social World*, S. D. Reese, O. H. Gandy, Jr., and A. E. Grant, Eds. Hillsdale, NJ, USA: Lawrence Erlbaum, 2001, pp. 231–242.
- [53] J. L. Himmelstein, "How the mass media use numbers to tell a story: The case of the crack scare of 1986," *Numeracy*, vol. 7, no. 1, 2014, Art. no. 2, doi: [10.5038/1936-4660.7.1.2](https://doi.org/10.5038/1936-4660.7.1.2).
- [54] Cision Media Research, "Top ten U.S. daily newspapers," Jan. 4, 2019.
- [55] Opioid Epidemic, *The New York Times*, 2020. [Online]. Available: <https://www.nytimes.com/spotlight/opioid-epidemic>
- [56] P. Kerr, "Anatomy of the drug issue: How, after years, it erupted," *The New York Times*, p. 1, 28, Nov. 17, 1986. [Online]. Available: <https://www.nytimes.com/1986/11/17/us/anatomy-of-the-drug-issue-how-after-years-it-erupted.html>
- [57] C. Reinerman and H. G. Levine, Eds., *Crack in America: Demon Drugs and Social Justice*. Oakland, CA, USA: Univ. California Press, 1997.
- [58] Department of Justice, Office of the Inspector General. (1997). The CIA-Contra Crack Cocaine Controversy: A review of the justice department's investigations and prosecutions. [Online]. Available: <https://fas.org/irp/agency/doj/oig/c4rpt/ch06p2.htm>
- [59] D. Kopf, "When did charts become popular?" *Priceonomics*, Sep. 14, 2016. [Online]. Available: <https://priceonomics.com/when-did-charts-become-popular>
- [60] D. Freed, "Crime: Justice system suffering from handling overload of cases," *The Los Angeles Times*, pp. A1, A40, Dec. 16, 1990.
- [61] L. Bernstein and C. Ingraham, "Fueled by drug crisis, U.S. life expectancy declines for a second straight year," *The Washington Post*, Dec. 21, 2017. [Online]. Available: https://www.washingtonpost.com/national/health-science/fueled-by-drug-crisis-us-life-expectancy-declines-for-a-second-straight-year/2017/12/20/2e3f8dea-e596-11e7-ab50-621fe0588340_story.html?utm_term=.55b2779eafd9
- [62] B. Farr and C. McGraw, "Public fear a necessary ally: Drug enforcers getting bombed in cocaine war," *The Los Angeles Times*, pp. 1, 3, 28, Sep. 21, 1986.
- [63] L. Perini, "The truth in pictures," *Philos. Sci.*, vol. 72, no. 1, pp. 262–285, 2005, doi: [10.1086/426852](https://doi.org/10.1086/426852).
- [64] A. Emery, *Depict Data Studio*, E-book, 2018. [Online]. Available: <https://depictdatastudio.com/great-graphs-ebook/>
- [65] B. G. Glaser and A. L. Strauss, *The Discovery of Grounded Theory: Strategies for Qualitative Research*. New Brunswick, NJ, USA: Aldine Transaction, 1999.
- [66] K. Charmaz, *Constructing Grounded Theory: A Practical Guide Through Qualitative Analysis*, 2nd ed. Thousand Oaks, CA, USA: Sage, 2014.
- [67] T. C. Portewig, "The role of rhetorical invention for visuals: A qualitative study of technical communicators in the workplace," *Tech. Commun.*, vol. 55, no. 4, pp. 333–342, 2008.
- [68] C. Teston, "Moving from artifact to action: A grounded investigation of visual displays of evidence during medical deliberations," *Tech. Commun. Quart.*, vol. 21, pp. 187–209, 2012, doi: [10.1080/10572252.2012.650621](https://doi.org/10.1080/10572252.2012.650621).
- [69] J. W. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 4th ed. Thousand Oaks, CA, USA: Sage, 2014.
- [70] V. Churchville, "Drug gripping youths at age 13, report finds," *The Washington Post*, pp. C1–C2, Jun. 10, 1988.
- [71] P. Bump, "Murder and crime get Trump's attention. Drug overdoses kill far more people," *The Washington Post*, Oct. 16, 2017. [Online]. Available: https://www.washingtonpost.com/news/politics/wp/2017/10/16/murder-and-crime-gets-trumps-attention-drug-overdoses-kill-far-more-people/?utm_term=.f5b499d24ce8
- [72] L. Bernstein and S. Higham, "Investigation: The DEA slowed enforcement while the opioid epidemic grew out of control," *The Washington Post*, Oct. 22, 2016. [Online]. Available: https://www.washingtonpost.com/investigations/the-dea-slowed-enforcement-while-the-opioid-epidemic-grew-out-of-control/2016/10/22/aea2bf8e-7f71-11e6-8d13-d7c704ef9fd9_story.html?utm_term=.ef366d8568cf
- [73] M. S. Greene, "'Sitting on a time bomb waiting for kids to die'; D.C. child welfare services overmatched," *The Washington Post*, pp. A1, A14, Sep. 12, 1989.
- [74] Editorial Board, "Young victims of the opioid epidemic," *The New York Times*, Jan. 16, 2017. [Online]. Available: <https://www.nytimes.com/2017/01/16/opinion/young-victims-of-the-opioid-epidemic.html>
- [75] S. Ungar, "Hot crises and media reassurance: A comparison of emerging diseases and Ebola Zaire," *Brit. J. Sociol.*, vol. 49, no. 1, pp. 36–56, 1998, doi: [10.2307/591262](https://doi.org/10.2307/591262).
- [76] R. L. Berke, "Capital offers a ripe market to drug dealers," *The New York Times*, pp. A1, A16, Mar. 28, 1989. [Online]. Available: <https://www.nytimes.com/1989/03/28/us/capital-offers-a-ripe-market-to-drug-dealers.html>
- [77] N. N. Levery, "Health repeal imperils opioid struggle," *The Los Angeles Times*, pp. A1, A8, Jun. 21, 2016.
- [78] J. Katz, "Short answers to hard questions about the opioid crisis," *The New York Times*, Aug. 10, 2017. [Online]. Available: <https://www.nytimes.com/interactive/2017/08/03/upshot/opioid-drug-overdose-epidemic.html>
- [79] L. Wheeler and S. Horwitz, "Operation clean sweep's future uncertain," *The Washington Post*, pp. A1, A14, Jan. 26, 1988.
- [80] H. Park and M. Bloch, "How the epidemic of drug overdose deaths rippled across America," *The New York Times*, Jan. 19, 2016. [Online]. Available: <https://www.nytimes.com/interactive/2016/01/07/us/drug-overdose-deaths-in-the-us.html>

- [81] J. Guo, "What Donald Trump and dying white people have in common," *The Washington Post*, Dec. 15, 2015. [Online]. Available: https://www.washingtonpost.com/news/wonk/wp/2015/12/15/what-donald-trump-and-dying-white-people-have-in-common-2/?utm_term=.21830a16ed33
- [82] P. Kerr, "Addiction's hidden toll: Poor families in turmoil," *The New York Times*, pp. 1, 28, Jun. 23, 1988. [Online]. Available: <https://www.nytimes.com/1988/06/23/nyregion/addiction-s-hidden-toll-poor-families-in-turmoil.html>
- [83] Editorial Board, "Crack; A disaster of historic dimension, still growing," *The New York Times*, p. 133, May 28, 1989. [Online]. Available: <https://www.nytimes.com/1989/05/28/opinion/crack-a-disaster-of-historic-dimension-still-growing.html>
- [84] J. B. Treaster, "U.S. cocaine epidemic shows signs of waning," *The New York Times*, p. 14, Jul. 1, 1990. [Online]. Available: <https://www.nytimes.com/1990/07/01/us/us-cocaine-epidemic-shows-signs-of-waning.html>
- [85] J. B. Treaster, "New York State reports a drop in crack traffic," *The New York Times*, p. 25, Dec. 27, 1990. [Online]. Available: <https://www.nytimes.com/1990/12/27/nyregion/new-york-state-reports-a-drop-in-crack-traffic.html>
- [86] J. Keller and A. Pearce, "A small Indiana county sends more people to prison than San Francisco and Durham, N.C., combined: Why?" *The New York Times*, Sep. 2, 2016. [Online]. Available: <https://www.nytimes.com/2016/09/02/upshot/new-geography-of-prisons.html>
- [87] H. Ryan, S. Glover, and L. Girion, "More than 1 million OxyContin pills ended up in the hands of criminals and addicts: What the drugmaker knew," *The Los Angeles Times*, pp. A1, A13–A15, Jul. 10, 2016.
- [88] J. Zacks, E. Levy, B. Tversky, and D. Schiano, "Graphs in print," in *Diagrammatic Representation and Reasoning*, M. Anderson, B. Meyer, and P. Olivier, Eds. London, UK: Springer-Verlag, 2002, pp. 187–206.
- [89] M. Friendly, "The golden age of statistical graphics," *Statist. Sci.*, vol. 23, pp. 502–535, 2008.
- [90] J. Berke and S. Gould, "Legal marijuana just went on sale in Illinois. Here are all the states where cannabis is legal," *Business Insider*, Jan. 2020. [Online]. Available: <https://www.businessinsider.com/legal-marijuana-states-2018-1>
- [91] C. Storrs, "Public health experts: Decriminalize minor drug offenses," CNN, Mar. 25, 2016. [Online]. Available: <https://www.cnn.com/2016/03/25/health/decriminalize-minor-drug-offenses-public-health/index.html>
- [92] K. Q. Seelye, "In annual speech, Vermont governor shifts focus to drug abuse," *The New York Times*, Jan. 8, 2014. [Online]. Available: <https://www.nytimes.com/2014/01/09/us/in-annual-speech-vermont-governor-shifts-focus-to-drug-abuse.html>
- [93] J. Capehart, "Chris Christie gets it on drug addiction," *The Washington Post*, Nov. 4, 2015. [Online]. Available: https://www.washingtonpost.com/blogs/post-partisan/wp/2015/11/04/chris-christie-gets-it-on-drug-addiction/?utm_term=.b010b858706d
- [94] J. Guo, "How dare you work on Whites': Professors under fire for research on White mortality," *The Washington Post*, Apr. 6, 2017. [Online]. Available: https://www.washingtonpost.com/news/wonk/wp/2017/04/06/how-dare-you-work-on-whites-professors-under-fire-for-research-on-white-mortality/?utm_term=.1a48d6e72013
- [95] A. R. Richards, "Argument and authority in the visual representations of science," *Tech. Commun. Quart.*, vol. 12, no. 2, pp. 183–206, 2003.
- [96] L. Walsh, "'Tricks,' hockey sticks, and the myth of natural inscription: How the visual rhetoric of Climategate conflated climate with character," in *Image Politics of Climate Change: Visualizations, Imaginations, Documentations*, B. Schneider and T. Nocke, Eds. Bielefeld, Germany: Transcript Verlag, 2014, pp. 81–104.
- [97] B. F. Barton and M. S. Barton, "Modes of power in technical and professional visuals," *J. Bus. Tech. Commun.*, vol. 7, no. 1, pp. 138–162, 1993.
- [98] J. Wolfe, "Teaching students to focus on the data in data visualization," *J. Bus. Tech. Commun.*, vol. 29, no. 3, pp. 344–359, 2015, doi: [10.1177/1050651915573944](https://doi.org/10.1177/1050651915573944).
- [99] J. Frith, "Big data, technical communication, and the smart city," *J. Bus. Tech. Commun.*, vol. 31, no. 2, pp. 168–187, 2017, doi: [10.1177/1050651916682285](https://doi.org/10.1177/1050651916682285).
- [100] C. Teston, L. Gonzales, K. Bivens, and K. Whitney, "Surveying precarious publics," *Rhetoric Health Med.*, vol. 2, no. 3, pp. 321–351, 2019, doi: [10.5744/rhm.2019.1015](https://doi.org/10.5744/rhm.2019.1015).
- [101] N. N. Jones and R. Walton, "Using narratives to foster critical thinking about diversity and social justice," in *Integrating Theoretical Frameworks for Teaching Technical Communication*, A. M. Haas and M. F. Eble, Eds. Logan, UT, USA: Utah State Univ. Press, 2018, pp. 241–267.
- [102] J. S. Colton and S. Holmes, "A social justice theory of active equality for technical communication," *J. Tech. Writing Commun.*, vol. 48, no. 1, pp. 4–30, 2018, doi: [10.1177/0047281616647803](https://doi.org/10.1177/0047281616647803).
- [103] P. Kerr, "Crack addiction: The tragic toll on women and their children," *The New York Times*, p. B2, Feb. 9, 1987. [Online]. Available: <https://www.nytimes.com/1987/02/09/nyregion/crack-addiction-the-tragic-toll-on-women-and-their-children.html>
- [104] H. Ryan, L. Girion, and S. Glover, "You want a description of hell? Oxycontin's 12-hour problem," *The Los Angeles Times*, pp. A1, A16, A18, May 5, 2016.
- [105] R. Morin and J. Allen, "Are we shooting ourselves in the foot in the war on drugs?" *The Washington Post*, pp. C1, C4, Jun. 26, 1988.
- [106] C. W. Dugger, "Cocaine use and infant mortality decline together in New York City," *The New York Times*, pp. 1, 26, Apr. 20, 1991. [Online]. Available: <https://www.nytimes.com/1991/04/20/nyregion/cocaine-use-and-infant-mortality-decline-together-in-new-york-city.html>
- [107] S. Karlamangla, "In California, opioid crisis is less pervasive," *The Los Angeles Times*, pp. A1, A12, Oct. 28, 2017.

- [108] E. Schmitt, "On the corner: Reign of crack and violence," *The New York Times*, pp. 37, 38, Oct. 13, 1987. [Online]. Available: <https://www.nytimes.com/1987/10/13/nyregion/on-the-corner-reign-of-crack-and-violence.html>
- [109] C. Ingraham, "Prescription painkillers are more widely used than tobacco, new federal study finds," *The Washington Post*, Sep. 20, 2016. [Online]. Available: https://www.washingtonpost.com/news/wonk/wp/2016/09/20/prescription-painkillers-are-more-widely-used-than-tobacco-new-federal-study-finds/?utm_term=.dd89c9aedad3
- [110] N. Irwin and B. Quoctrung, "The rich live longer everywhere: For the poor, geography matters," *The New York Times*, Apr. 11, 2016. [Online]. Available: <https://www.nytimes.com/interactive/2016/04/11/upshot/for-the-poor-geography-is-life-and-death.html>
- [111] A. G. Gross, "Medical tables, graphics and photographs: How they work," *J. Tech. Writing Commun.*, vol. 37, no. 4, pp. 419–433, 2007.
- [112] I. Wilkerson, "Crack and gunfire come to a quiet place," *The New York Times*, p. 10, Jan. 9, 1989. [Online]. Available: <https://www.nytimes.com/1989/01/09/us/crack-and-gunfire-come-to-a-quiet-place.html>
- [113] J. Katz, "Drug deaths in America are rising faster than ever," *The New York Times*, Jun. 5, 2017. [Online]. Available: <https://www.nytimes.com/interactive/2017/06/05/upshot/opioid-epidemic-drug-overdose-deaths-are-rising-faster-than-ever.html>
- [114] D. Freed, "A war of skirmishes," *The Los Angeles Times*, p. A33, Dec. 18, 1990.
- [115] S. G. Boodman, "Gonorrhea, syphilis rise, sharply here," *The Washington Post*, pp. A1, A15, Apr. 6, 1989.
- [116] L. Sun, "Rural Americans are more likely to die from the top 5 causes of death," *The Washington Post*, Jan. 12, 2017. [Online]. Available: https://www.washingtonpost.com/news/to-your-health/wp/2017/01/12/rural-americans-more-likely-to-die-from-top-5-causes-of-death/?utm_term=.8d36419b155a
- [117] A. Cha, "The drug industry's answer to opioid addiction: More pills," *The Washington Post*, Oct. 16, 2016. [Online]. Available: https://www.washingtonpost.com/national/the-drug-industrys-answer-to-opioid-addiction-more-pills/2016/10/15/181a529c-8ae4-11e6-bff0-d53f592f176e_story.html?utm_term=.342b97ac16ac
- [118] R. L. Berke, "Survey shows use of drugs by students fell last year," *The New York Times*, p. A16, Feb. 14, 1990. [Online]. Available: <https://www.nytimes.com/1990/02/14/us/survey-shows-use-of-drugs-by-students-fell-last-year.html>
- [119] K. Humphreys, "Where people drink the most booze and do the most drugs," *The Washington Post*, May 23, 2016. [Online]. Available: https://www.washingtonpost.com/news/wonk/wp/2016/05/23/where-people-drink-the-most-booze-and-do-the-most-drugs/?utm_term=.4f2b4a2409ab
- [120] A. Goodnough and S. Tavernise, "Opioid prescriptions drop for first time in two decades," *The New York Times*, May 20, 2016. [Online]. Available: <https://www.nytimes.com/2016/05/21/health/opioid-prescriptions-drop-for-first-time-in-two-decades.html>
- [121] N. Lewis, "Deadliest overdose month in D.C. setting record pace for year," *The Washington Post*, pp. D1, D5, Apr. 26, 1988.
- [122] P. W. Valentine, "D.C., Baltimore worlds apart in homicide rates," *The Washington Post*, pp. D1, D7, Apr. 16, 1990.
- [123] C. Ingraham, "Heroin deaths surpass gun homicides for the first time, CDC data shows," *The Washington Post*, Dec. 8, 2016. [Online]. Available: <https://www.washingtonpost.com/news/wonk/wp/2016/12/08/heroin-deaths-surpass-gun-homicides-for-the-first-time-cdc-data-show>

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